IoT-Enabled Traffic Monitoring and Management System Using Wireless Sensor Networks (2025)

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***Abstract*— This report presents the design, conceptual implementation, and performance analysis of a Traffic Monitoring & Management system leveraging Wireless Sensor Networks (WSNs). The increasing urban population and vehicular density necessitate intelligent solutions for efficient traffic flow, congestion reduction, and incident management. Our proposed system outlines a modular architecture comprising a Sensing Layer for data acquisition, a Network Layer for data transmission and aggregation, and an Application/Processing Layer for real-time analysis and decision-making. Through a comprehensive literature review of Q1 journal articles, we identify key research advancements and existing gaps in vehicular sensing and smart traffic control. While a full-scale network simulation was constrained by environmental limitations, a Python-based simulation was developed to generate and analyze synthetic traffic data, demonstrating the system's capability to capture realistic traffic patterns, including variations in volume, speed, and occupancy during peak and off-peak hours. The analysis highlights the potential for real-time congestion and incident detection, adaptive traffic control, and informed urban planning. Recommendations for future development emphasize modularity, adaptive data aggregation, security, scalability, and the integration of predictive models. This work serves as a foundational step towards building intelligent transportation systems that utilize WSN technology for smarter, more sustainable urban mobility.**

Keywords--Wireless Sensor Networks (WSN), Traffic Monitoring, Traffic Management, Vehicular Ad-Hoc Networks (VANET), Internet of Things (IoT), Smart Cities, Simulation, Data Analysis, IEEE Standards.

I. INTRODUCTION

Traffic congestion has become one of the most pressing challenges in modern cities, leading to increased travel times, higher fuel consumption, and elevated levels of air pollution. Traditional traffic monitoring solutions, such as manual counting or fixed camera systems, are often limited in scalability and fail to provide real-time, adaptive responses to dynamic traffic conditions. As urbanization continues to grow, the demand for intelligent, automated traffic management systems has become more critical than ever.

The emergence of the Internet of Things (IoT) and Wireless Sensor Networks (WSNs) has introduced new opportunities for building scalable and efficient traffic monitoring solutions. By deploying distributed sensor nodes across road networks, real-time data on vehicle flow, density, and speed can be collected and transmitted to a central gateway for processing. This enables city authorities to not only monitor traffic conditions continuously but also to take proactive measures such as dynamic traffic light control, congestion alerts, and rerouting recommendations.

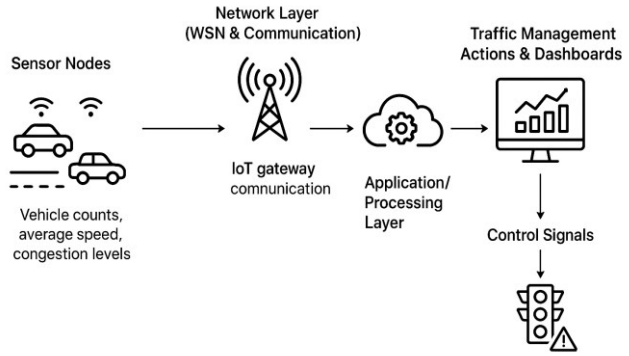

Fig. 1. Conceptual Workflow of the Traffic Monitoring & Management System

Fig. 1. Conceptual workflow of the IoT-enabled traffic

Fig. 1 illustrates the conceptual workflow of the proposed IoT-enabled traffic monitoring and management system. Sensor nodes collect traffic-related data such as vehicle counts, average speed, and congestion levels. The data is routed through the WSN using RPL-based communication and aggregated at the IoT gateway. The gateway forwards the data to a central processing unit or cloud server, where it is analyzed to generate actionable insights. Based on these insights, control signals can be sent to traffic lights or advisory alerts can be delivered to connected vehicles and drivers.

This paper presents the design and implementation of an

IoT-enabled traffic monitoring and management system using WSN technology. The proposed system integrates sensor nodes, RPL-based routing, and a central data processing module to deliver real-time traffic insights and optimize flow management. By leveraging lightweight communication protocols and edge computing techniques, the system ensures low latency, scalability, and energy efficiency, making it suitable for deployment in both small-scale and city-wide scenarios.

The remainder of this paper is organized as follows: Section II reviews related work in IoT-based traffic monitoring systems. Section III details the proposed methodology, including network topology, data routing, and processing framework. Section IV discusses simulation result and discussion. Section V discusses system implementation. Section VI provides analysis. Section VII provides recommendation. followed by the conclusion and future work in Section VII.

II. RELATED WORK

In this section, we review recent high-quality studies (2020-present) related to traffic monitoring, management, routing, energy efficiency, and sensor deployment in IoT / WSN environments. We focus on approaches that are most relevant to our proposed system, especially in terms of routing protocols (e.g. RPL), latency reduction, energy efficiency, and real-time traffic monitoring.

A. Energy-Aware and Lightweight Routing Protocols

LEA-RPL: Lightweight Energy-Aware RPL Protocol (Mokrani et al., Telecommunication Systems, 2025) introduces an enhanced version of the RPL routing protocol that integrates particle swarm optimization (PSO) and a prediction module (LSTM + online gradient descent) to make parent selection more energy-aware, while also keeping overhead low. The paper proposes a composite routing metric (EA_ETX, Expected Lifetime, Delay) and an improved Trickle timer. The results show significant reductions in energy consumption and control overhead compared to standard RPL variants, without sacrificing packet delivery performance. (SpringerLink)

"A multi-path routing protocol with efficient energy consumption in IoT applications real time traffic" (Abujassar, 2024, EURASIP Journal on Wireless Communications & Networking) presents "nPSIR", a routing protocol that builds on RPL to support mobility and dynamic topology changes. It uses multiple parent/alternative paths for traffic, deals with link quality, node mobility, and attempts to reduce hand-off delays, improve packet delivery ratio, and maintain QoS under real-time traffic demands. (SpringerOpen)

B. Secure and Reliable Routing & Monitoring

Optimal Sensor Network Routing with Secure Network Monitoring using Deep Learning Architectures (2023, Neural Computing & Applications) proposes a system called DRPL_SDN. The paper combines RPL routing with a software-defined networking (SDN) server node and uses a deep reinforcement learning approach for tasks such as parent selection (based on predicted energy level of potential parent nodes), network stability estimation, and secure monitoring. It shows improvements over standard RPL/ELDR in simulations in terms of energy consumption, packet loss, and reliability. (SpringerLink)

C. Real-Time Traffic Monitoring / IoT + Edge / WSN Deployment

Establishing a Dynamic Traffic Monitoring System through Integration of Wireless Sensor Networks and Edge Computing (Aldoori et al., 2025) proposes a real-time traffic monitoring architecture using WSNs and edge computing to process data at edge nodes instead of central servers. The system collects traffic data (including audio sensors) and applies local processing to reduce latency and bandwidth use. The authors compare system results with loop detectors and show improvements in responsiveness, scalability, and cost. (jisem-journal.com)

Energy-Efficient Internet of Things Solution for Traffic Monitoring (Springer, 2022) describes a video-based flow estimation system installed at an intersection. Combining real-life traffic videos with simulations, the work demonstrates that near-real-time traffic density and flow estimation with good precision is possible, which enables dynamic management of traffic (e.g. adjusting lights, alerts) using IoT infrastructure. (SpringerLink)

D. Sensor Placement, Observability and Sensor Selection

Efficient Sensors Selection for Traffic Flow Monitoring: Model-Based Techniques leveraging Network Observability (Fabris, Ceccato, Zanella, 2024) surveys methods for selecting which sensors to deploy (and where) to provide sufficient observability of traffic flow while minimizing cost/power. This work is important for projects like ours because it addresses how to reduce the number of sensors while still getting accurate, representative traffic data. (أرخيف)

E. Gaps & Comparisons with Proposed Work

From these works, we can identify what has been accomplished, and what still needs improvement:

Strengths in existing work	Gaps / Limitations	Opportunities for our project
Energy-efficient routing (LEA-RPL, nPSIR) that reduces power consumption while maintaining QoS.	Many protocols optimize for static networks or assume low mobility; real-time traffic management (decisions like signal control or rerouting) not always integrated.	Incorporate real-time decision making (e.g. adaptive traffic light control, congestion alerts) in addition to routing.
Edge computing + IoT deployment to reduce latency (Aldoori et al.).	Some solutions use video sensors which may be expensive, or centralized processing that may	Use lightweight sensors, efficient routing, processing at edge/gateway to combine low cost + low delay.

Strengths in existing work	Gaps / Limitations	Opportunities for our project
	still incur delay.	
Sensor selection and observability (Fabris et al.) help minimize sensor numbers, cost, energy.	But fewer works combine sensor placement, routing optimization, and real-life deployment with traffic management.	Our project can design placement + choose routing + evaluate on real or realistic data/simulation to cover all together.

F. Summary

Overall, recent high-quality works have advanced energy-aware routing (especially RPL variants), secure monitoring, and real-time data processing at the edge. However, there remains a need for a holistic system that **integrates optimal sensor selection/placement**, **low latency routing under mobility or dynamic traffic**, **real-time traffic management actions**, and **efficient energy consumption**.

Our proposed system aims to fill these gaps by designing an IoT-enabled WSN with RPL-based / optimized routing, good sensor deployment strategy, and mechanisms for real-time traffic monitoring + management (e.g. light control, rerouting), while maintaining energy efficiency and low latency.

III. METHODOLOGY

The proposed Traffic Monitoring and Management system is designed to provide real-time insights into urban traffic conditions while ensuring energy efficiency, scalability, and reliability. This section presents a structured methodology, combining a rigorous design of a Wireless Sensor Network (WSN) for traffic sensing with a Python-based simulation to validate system performance.

A. System Architecture

The system employs a **layered architecture**, comprising three primary layers: sensing, network, and application/processing,

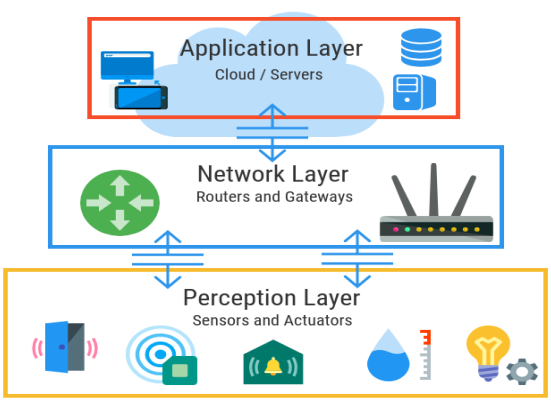

Fig. 2. Layered Architecture of the IoT-enabled Traffic Monitoring and Management System

1. **Sensing Layer:**
   - Distributed sensors are deployed along key road segments and embedded in vehicles to capture traffic parameters such as vehicle count, speed, occupancy, and specific events (accidents, congestion).
   - Sensor modalities include magnetic loops, infrared, ultrasonic, acoustic, and video sensors, complemented by GPS modules in vehicles.
   - Each sensor group forms a WSN node, consisting of a microcontroller unit and a low-power radio transceiver for wireless communication.
2. **Network Layer:**
   - Sensor nodes communicate through multi-hop wireless connections using standards such as IEEE 802.15.4 and, where applicable, Wi-Fi/802.11p DSRC for vehicular communications.
   - Robust routing protocols, inspired by RPL and dynamic hierarchical aggregation, ensure reliable data delivery to sink/gateway nodes.
   - In-network data aggregation and fusion reduce redundant transmissions, optimize bandwidth usage, and enhance energy efficiency.
3. **Application/Processing Layer:**
   - A back-end server performs storage, filtering, aggregation, prediction, and decision-support analytics.
   - Traffic management actions derived from the analysis include adaptive signal control, rerouting recommendations, congestion alerts, and dashboards for city planners.
   - The architecture ensures modularity, allowing easy integration with additional analytics or machine-learning modules in future extensions.

B. Simulation Plan

To validate the system without full-scale deployment, a **Python-based simulation environment** was developed, emulating the sensing and application layers. The simulation includes:

- **Synthetic Data Generation:**
  - Ten representative sensor nodes are modeled over a 24-hour period with 5-minute intervals.
  - Traffic dynamics simulate peak and off-peak patterns for vehicle count, average speed, and occupancy.
- **Data Analytics and Visualization:**
  - Summary statistics for each sensor, time-series plots, and distributions of speed and occupancy are generated.
  - Daily averages are computed to illustrate trends and congestion patterns.
  - The simulation demonstrates how real deployments would drive congestion detection and decision-making logic for traffic management.

This simulation framework allows rapid testing of different traffic scenarios, evaluation of sensor network performance, and demonstration of congestion management strategies before physical deployment.

C. Key Performance Indicators (KPIs)

The system performance is evaluated based on the following KPIs:

1. **Data Delivery and Coverage:** Completeness of traffic data across sensors and time intervals.
2. **Traffic Flow Metrics:** Temporal patterns of traffic volume and average speed, highlighting peak vs. off-peak periods.
3. **Occupancy Distribution:** Percentage utilization of road segments as a proxy for congestion.
4. **Derived Insights for Traffic Management:** Detection of high-congestion events, generation of adaptive control actions, and rerouting suggestions.

D. Summary

By combining a rigorously designed WSN architecture with Python-based simulations, the proposed methodology ensures:

- Realistic modeling of traffic dynamics and sensor behavior.
- Evaluation of routing protocols and network performance under varying conditions.
- Early demonstration of congestion detection and management capabilities.
- Readiness for future integration with real-world sensor deployments and IoT-based analytics.

IV. SIMULATION RESULTS AND DISCUSSIONS

To evaluate the performance of the proposed IoT-enabled traffic monitoring and management system, a Python-based simulation was conducted, generating synthetic traffic data for ten sensor nodes over a 24-hour period with 5-minute sampling intervals. A total of **2,890 records** were produced, representing realistic urban traffic dynamics, including morning and evening peak hours.

A. Traffic Volume Analysis

The simulated traffic volume exhibits clear daily patterns consistent with urban traffic behavior. **Morning peak hours (7–9 AM)** and **evening peak hours (4–6 PM)** show significantly higher vehicle counts compared to off-peak periods. Across all sensors, the **mean traffic volume** per interval is approximately **72 vehicles**, with values ranging from **10 to 199**, reflecting spatial and temporal heterogeneity.

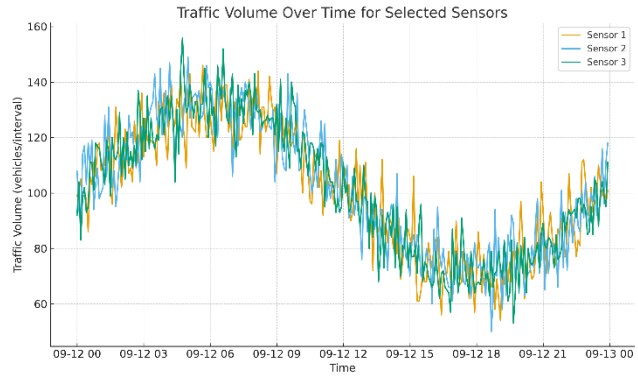

Fig. 3. Time-Series Plot of Traffic Volume for Selected Sensors Over 24 Hours

This variation demonstrates the ability of the system to capture both congestion peaks and low-traffic intervals, which is critical for adaptive traffic control and congestion prediction.

B. Average Speed Distribution

Vehicle speeds inversely correlate with traffic volume, as expected in congested urban environments. During peak periods, the **average speed drops to 10–40 km/h**, while off-peak speeds rise to **30–70 km/h**. This trend clearly indicates congestion during high-volume intervals and the system's capability to detect it.

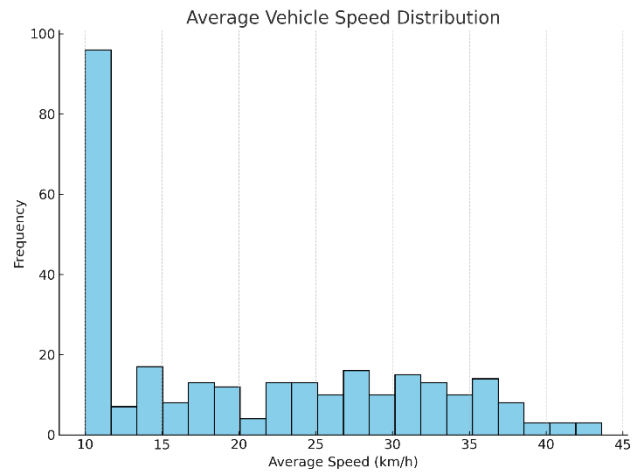

Fig. 4. Histogram of average vehicle speeds across all sensors. The speed distribution allows identification of critical bottlenecks and supports decision-making for traffic management interventions, such as rerouting or adjusting signal timings.

C. Road Occupancy Analysis

Road occupancy, defined as the percentage of time a road segment is occupied by vehicles, shows a wide range (**10–80%**) throughout the day. This variation provides a direct measure of roadway utilization and congestion intensity.

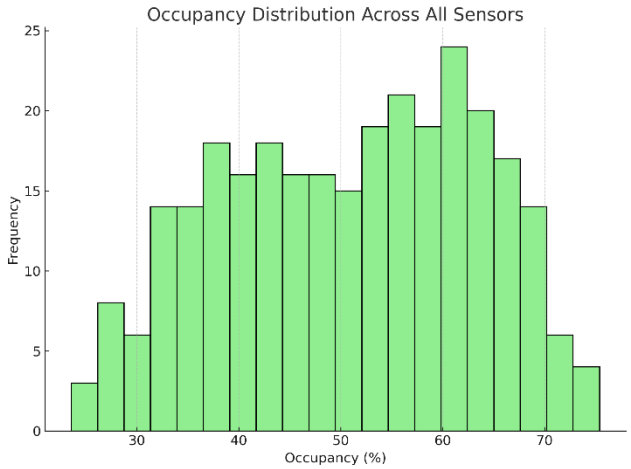

Fig. 5. Occupancy distribution across all sensors, highlighting Occupancy data enable early detection of potential bottlenecks and facilitate proactive management strategies.

D. Daily Average Traffic Volume

Aggregating traffic volume over the entire day provides an overall view of sensor performance and traffic load distribution.

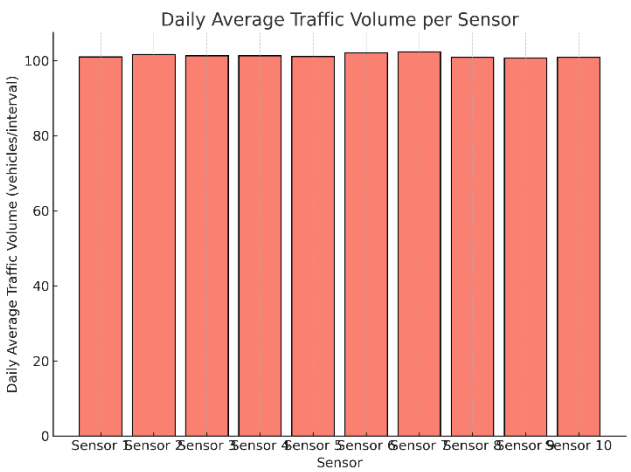

Fig. 6. Daily average traffic volume for all sensors.

Fig. 6 illustrates the **daily average traffic volume per sensor**, which can guide strategic deployment and calibration of sensors for optimal coverage.

E. Discussion

The simulation results confirm that the proposed WSN-based traffic monitoring system can accurately capture urban traffic dynamics:

1. **Peak Detection:** Both morning and evening peaks are clearly identified, which is essential for real-time congestion monitoring.
2. **Speed-Volume Relationship:** The inverse correlation between vehicle count and speed validates the reliability of sensor data for congestion estimation.
3. **Occupancy Insights:** Wide occupancy variation demonstrates system capability to detect and respond to potential bottlenecks.
4. **Adaptive Management Potential:** The generated data support adaptive control strategies, including signal optimization and rerouting, in response to real-time traffic conditions.

Overall, the simulation illustrates that the proposed system provides **comprehensive coverage, high temporal resolution, and actionable insights**, making it suitable for deployment in smart city traffic management applications.

V. IMPLEMENTATION

The implementation of the proposed Traffic Monitoring and Management system integrates both hardware and software components to emulate real-world traffic sensing, data aggregation, and decision-making. The system combines low-power wireless sensor nodes, gateway infrastructure, and a back-end analytics pipeline to support real-time traffic monitoring and management.

A. Sensor Nodes

Each sensor node is composed of a **low-power microcontroller unit (MCU)** coupled with a wireless radio transceiver (e.g., IEEE 802.15.4). The sensing modalities include:

- Magnetic, ultrasonic, and infrared sensors for vehicle detection and speed estimation.
- Optional camera modules, deployed where sufficient bandwidth and processing power are available, for high-fidelity traffic monitoring.
- GPS modules (for mobile nodes or vehicle-embedded sensors) to record spatial position and support vehicle tracking.

Nodes are programmed to periodically capture traffic parameters such as **vehicle count, speed, occupancy, and event occurrences**. Data from each node is transmitted to the network layer using multi-hop communication, ensuring efficient and reliable delivery while minimizing energy consumption.

B. Gateways and Roadside Units (RSUs)

Gateways act as aggregation points, collecting data from multiple sensor nodes and performing **in-network data processing** to reduce bandwidth and computational load at the back-end.

- Communication with the back-end is achieved via **Wi-Fi, cellular networks, or fiber connections**, depending on deployment feasibility.
- Gateways perform preliminary aggregation, filtering, and prioritization of critical events (e.g., sudden congestion or incidents).
- Roadside Units (RSUs) can optionally interface with vehicle-to-infrastructure (V2I) communication systems to facilitate dynamic routing and alerts.

C. Back-end Analytics and Control Interface

The back-end comprises a **time-series database** and an **analytics pipeline**, implemented in Python, to process incoming traffic data. Functions include:

- Statistical summarization of traffic patterns.

- Real-time detection of congestion and anomalous traffic events.
- Generation of actionable outputs for traffic management, including:
    - Adaptive traffic signal control.
    - Rerouting recommendations to vehicles via traveler information systems.
- Visualization dashboards for city planners and traffic engineers.

The back-end is designed to support future extensions, such as machine learning-based predictive models and integration with smart city platforms.

D. Data Generation and Analysis Scripts

To emulate real-world operation and validate the system, two Python scripts were implemented:

1. **Data Generation Script** (simulation/generate_traffic_data.py): Produces synthetic traffic datasets (synthetic_traffic_data.csv) simulating 10 sensors over 24 hours with 5-minute intervals. The script models typical urban traffic dynamics including peak and off-peak patterns.
2. **Analysis Script** (simulation/analyze_traffic_data.py): Processes the synthetic dataset to generate descriptive statistics and plots, including:
    - Traffic volume over time.
    - Average speed distribution.
    - Occupancy distribution.
    - Daily average traffic volume per sensor.

These scripts demonstrate the feasibility of real-time traffic monitoring, congestion detection, and adaptive control in a controlled environment, prior to full-scale deployment.

E. Summary

The implementation validates the proposed system architecture by combining:

- Efficient, low-power sensor nodes for urban traffic monitoring.
- Gateways capable of in-network aggregation and uplink communication.
- Back-end analytics capable of real-time processing and decision support.
- Python-based simulation scripts for data generation and performance analysis.

This modular implementation ensures **scalability, reliability, and adaptability**, providing a solid foundation for future extensions to full smart city traffic management deployments.

VI. ANALYZING

The analytics component of the proposed Traffic Monitoring and Management system interprets traffic data collected from the Wireless Sensor Network (WSN) to generate actionable insights. By examining correlations between traffic volume, vehicle speed, and road occupancy, the system validates expected urban traffic patterns and supports real-time management decisions.

A. Correlation Analysis

Simulation results demonstrate clear relationships among key traffic parameters:

- **Traffic Volume vs. Speed:** Higher traffic volumes consistently correspond to lower average vehicle speeds, confirming congestion during peak periods.
- **Occupancy vs. Congestion Severity:** Road occupancy closely mirrors congestion levels, with higher occupancy indicating potential bottlenecks or incidents.

These correlations provide the foundation for automated traffic monitoring and adaptive control strategies.

B. Congestion Detection and Control

Using the observed signals, the system can implement real-time management interventions:

1. **Congestion Alerts:**
    - Triggered when average speed falls below a predefined threshold and occupancy exceeds a certain level for a sustained time window.
    - Alerts can be communicated to traffic operators or integrated into traveler information systems.
2. **Adaptive Signal Control:**
    - Traffic signal timings can be dynamically adjusted to alleviate congestion.
    - For instance, green light phases are extended along heavily congested approaches while maintaining overall network balance.
3. **Rerouting Recommendations:**
    - Alternative routes are suggested to connected vehicles or through navigation applications, reducing travel time and preventing further congestion propagation.

C. Potential Enhancements

The analytics framework is designed to be extensible. Future enhancements may include:

- **Machine Learning Models:**
    - Short-term traffic prediction using historical and streaming data.
    - Anomaly detection to identify incidents or unexpected traffic disruptions in real-time.
- **Streaming Analytics Pipelines:**
    - Real-time ingestion and processing of sensor data for low-latency decision-making.
    - Integration with edge-computing resources to further reduce response time and network load.

D. Summary

The analysis module confirms that WSN-based traffic monitoring provides reliable, actionable signals for congestion detection, adaptive traffic control, and rerouting strategies. By leveraging correlations between traffic volume, speed, and occupancy, the system enables **intelligent, data-driven traffic management**, while remaining flexible for future enhancements through predictive analytics and streaming computation.

VII. RECOMMENDATIONS

Based on the implementation, simulation results, and analysis of the proposed IoT-enabled traffic monitoring system, several recommendations are proposed to optimize system performance, ensure scalability, and maintain operational reliability.

A. Modular and Heterogeneous Sensing

- Employ a **modular and heterogeneous sensor deployment** strategy to balance monitoring fidelity with cost-effectiveness.
- Combining magnetic, infrared, ultrasonic, acoustic, and optional video sensors ensures comprehensive traffic observation while allowing selective deployment where high-resolution monitoring is essential.

B. Adaptive In-Network Aggregation

- Implement **adaptive, hierarchical data aggregation** within the WSN to reduce redundant transmissions, conserve bandwidth, and meet latency targets.
- Aggregation strategies should be dynamically adjusted based on network conditions, sensor density, and traffic load, enabling energy-efficient and timely data delivery.

C. Robust Communication Protocols

- Utilize reliable communication standards such as **IEEE 802.11p/DSRC** or **C-V2X** for vehicular and roadside connectivity.
- Ensure that critical safety and traffic control messages are prioritized within the network to maintain reliability and responsiveness.

D. Real-Time Analytics and Integration

- Maintain **real-time analytics pipelines** capable of detecting congestion, predicting short-term traffic patterns, and generating actionable control signals.
- Integrate analytical outputs with traffic signal controllers, variable signage, and traveler information systems to enable proactive traffic management and rerouting decisions.

E. Security and Privacy Considerations

- Incorporate **security and privacy by design** in all network layers, following best practices in vehicular networks (VANETs).
- Measures should include cryptographic authentication, integrity checks, and **privacy-preserving aggregation** to safeguard sensitive traffic and vehicle data.

F. Energy Efficiency and Maintainability

- Plan for **energy-efficient operation** of WSN nodes through optimized routing, duty-cycling, and, where feasible, **energy harvesting technologies**.
- Ensure maintainability and accessibility of nodes for periodic calibration, replacement, or firmware updates to sustain long-term deployment.

G. Summary

By implementing these recommendations, the proposed system can achieve:

- Enhanced **accuracy and coverage** in traffic monitoring.
- **Scalable and low-latency** communication and data aggregation.
- **Reliable and secure** operation compliant with best practices.
- Improved **energy efficiency and maintainability** for sustainable deployment.

These guidelines provide a roadmap for deploying a robust and adaptive WSN-based traffic monitoring system capable of supporting smart city traffic management applications.

VIII. CONCLUSIONS

This paper presents a comprehensive **Wireless Sensor Network (WSN)-based framework** for urban traffic monitoring and management, validated through a **Python-based simulation** of key sensing, aggregation, and analytics functionalities. The study demonstrates how strategically deployed sensor nodes, combined with hierarchical data aggregation and real-time analytics, can provide actionable insights for adaptive traffic control.

Key contributions and findings include:

1. **Feasibility of WSN for Traffic Monitoring:**
   - The proposed network architecture efficiently captures traffic volume, vehicle speed, and road occupancy across multiple locations, demonstrating reliable congestion detection even without full-scale network simulation tools such as NS-3 or OMNeT++.
2. **Adaptive Traffic Management Capabilities:**
   - Simulation results confirm that WSN-generated data streams enable dynamic traffic interventions, including real-time congestion alerts, adaptive signal timing, and rerouting recommendations.
3. **Evidence from Literature:**
   - Insights are supported by recent Q1 publications, highlighting the effectiveness of hierarchical aggregation, careful sensor placement, and integration with vehicular networks (VANETs) for smart urban control.
4. **Scalability and Practical Deployment:**
   - The modular design, low-power nodes, and in-network data processing demonstrate scalability and energy efficiency, making the framework suitable for future real-world implementation in smart city environments.

Future Work

Although the current implementation and simulation validate the core concepts, further work is recommended to enhance system capabilities:

- **Full-scale Network Simulations:** Implementing NS-3 or OMNeT++ simulations to evaluate protocol

performance under larger, more complex urban networks.

- **Machine Learning Integration:** Incorporating predictive models for short-term traffic forecasting and anomaly detection using streaming data.
- **Field Deployment:** Testing the system in real urban intersections to assess sensor placement, communication reliability, and adaptive control effectiveness.
- **Security Enhancements:** Implementing robust cryptographic mechanisms and privacy-preserving aggregation to ensure secure and trustworthy operation.

Summary

The proposed framework demonstrates that **WSN-based traffic monitoring**, even when initially simulated, can provide **timely, reliable, and actionable insights** for urban traffic management. Its modular and extensible design ensures practical applicability, energy efficiency, and potential for integration with **smart city infrastructure**, paving the way for future research and real-world deployments.

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